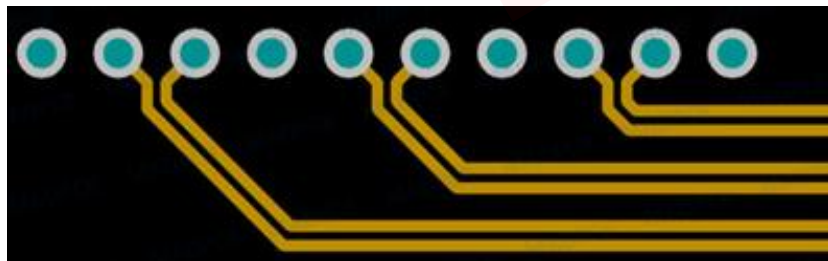
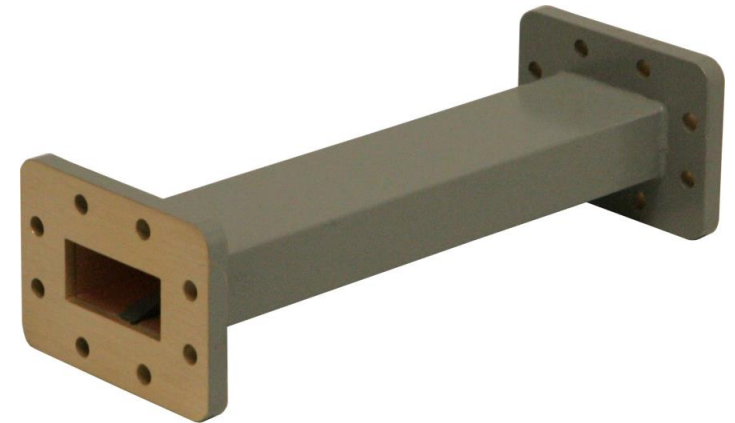
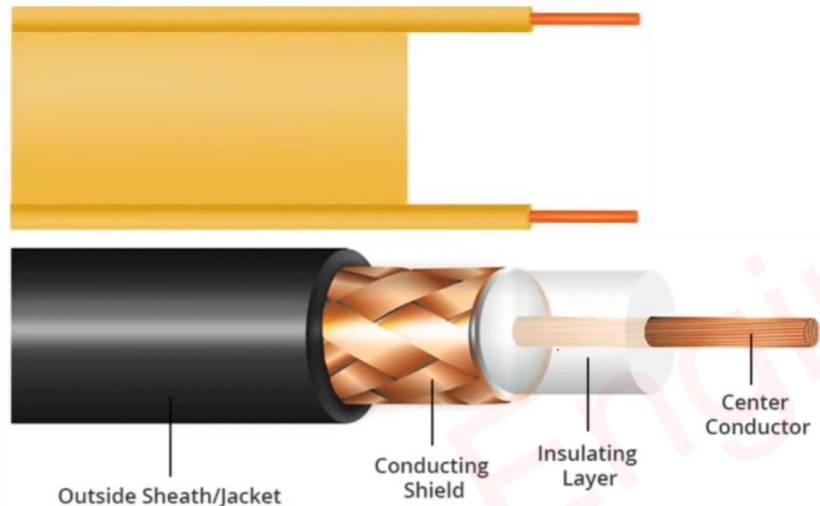


Tx Lines Vs Waveguides

Parameters	Transmission Line	Waveguide
Types	❖ Two Parallel Wire [In Telephone N/W] ❖ Coaxial Wire [In TV Signals] ❖ Microstrip Transmission Line [In PCB]	❖ Rectangular Waveguide ❖ Circular Waveguide
Metal used	❖ In Transmission Line, Two parallel metals are used.	❖ In Waveguide, a Hollow Metallic Tube is used.

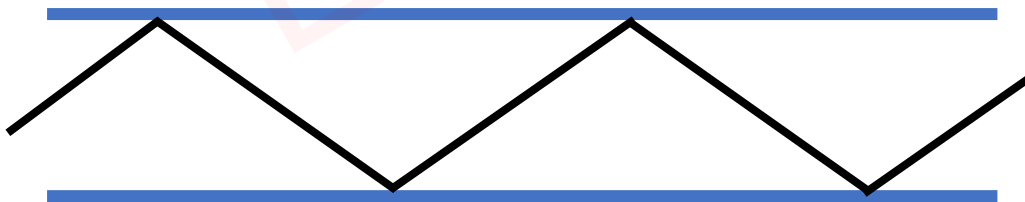
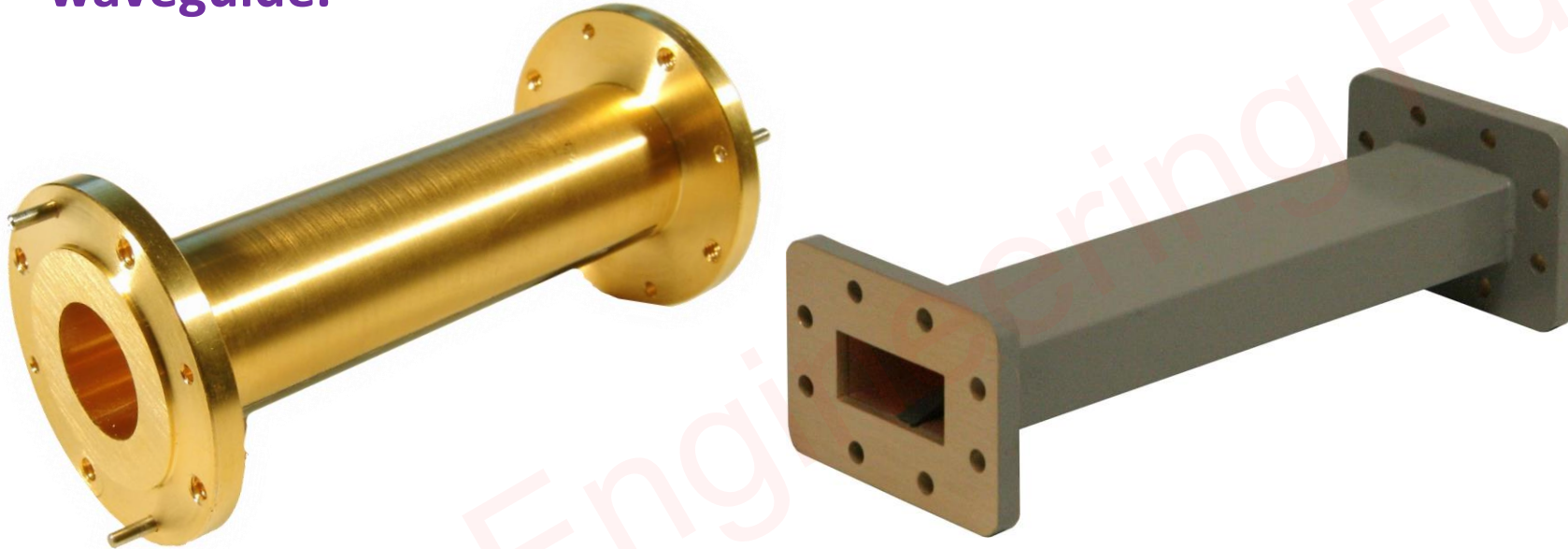


Parameters	Transmission Line	Waveguide
Modes of Propagation	❖ Transmission Line supports TEM Mode of Propagation. ❖ For Z direction of Propagation, In TEM Mode [$H_z = 0, E_z = 0$]	❖ Waveguide Only supports TE and TM modes of Propagation. ❖ For Z direction of Propagation, In TE Mode [$H_z \neq 0, E_z = 0$] In TM Mode [$H_z = 0, E_z \neq 0$]
Allowed Frequencies	❖ All Frequencies can be transmitted.	❖ It allows only those frequencies, which are greater than cut off frequency.
Signal Propagation	❖ Signal Couples between two wires. ❖ Signal Propagates based on Impedance matching at load.	❖ Signal Propagates based on TIR (Total Internal Reflection) from the walls of Waveguide.
Impedance	❖ It has Characteristics Impedance.	❖ It has wave impedance.
Theory	❖ Propagation of waves is according to "Circuit Theory"	❖ Propagation of waves is according to "Field Theory"
Earthing	❖ Return Conductor to earth is existed.	❖ There is no return conductor because the body of the waveguide acts as earth.
Bandwidth	❖ Bandwidth is not Limited	❖ Bandwidth is Limited.
Attenuation	❖ Signal Attenuation is high due to dielectric material loss.	❖ Signal Attenuation is low.
Power handling	❖ Suitable for Low transmitting power.	❖ Waveguide can handle high power.

Fundamentals of Waveguides

Basics of Waveguides

- ❑ Waveguide is a Hollow Metallic tube with a uniform cross-section.
- ❑ Waveguide is used to transmit EM wave using successive reflections from inner walls of waveguide.



Side view of Waveguide

EM waves propagation
Based on TIR

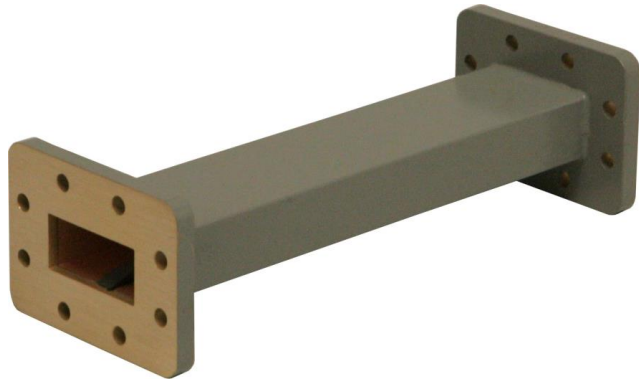
Advantages of Waveguides

- ☐ Waveguides are Easy to Manufacture.
- ☐ Waveguides have Immunity to EM wave interference.
- ☐ Waveguides are durable and reliable with their structure.
- ☐ Waveguides can support different modes of Propagation.
- ☐ Waveguides can Handle very high Power.
- ☐ In Waveguides, Loss of Power is also Low.
- ☐ In Waveguides, Attenuation is also Low.
- ☐ Waveguides have Lower Losses compared to Coaxial Cable.

Types of Waveguides

□ In general, there are five types of Microwave Waveguides:

- Rectangular Waveguide
- Circular Waveguide
- Elliptical Waveguide
- Single Ridged Waveguide
- Double Ridged Waveguide



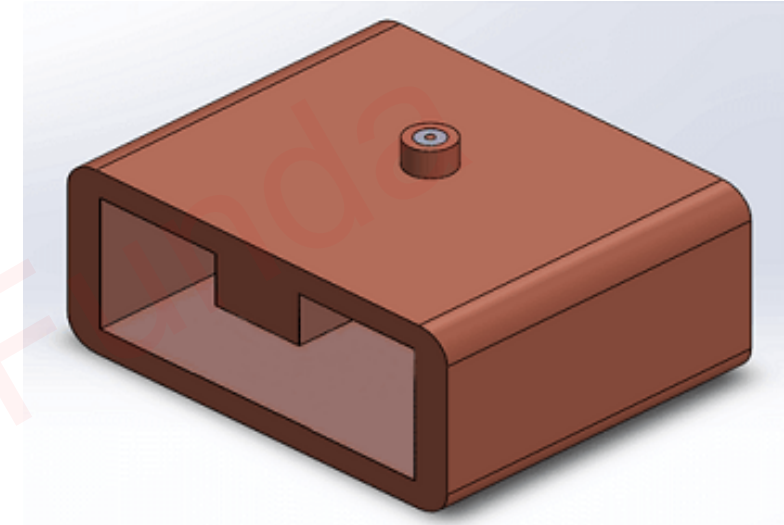
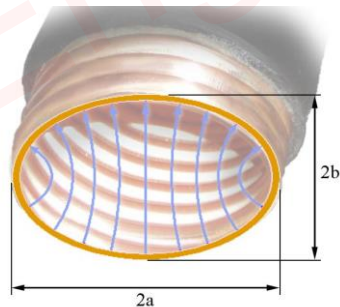
Rectangular Waveguide



Circular Waveguide



Elliptical Waveguide

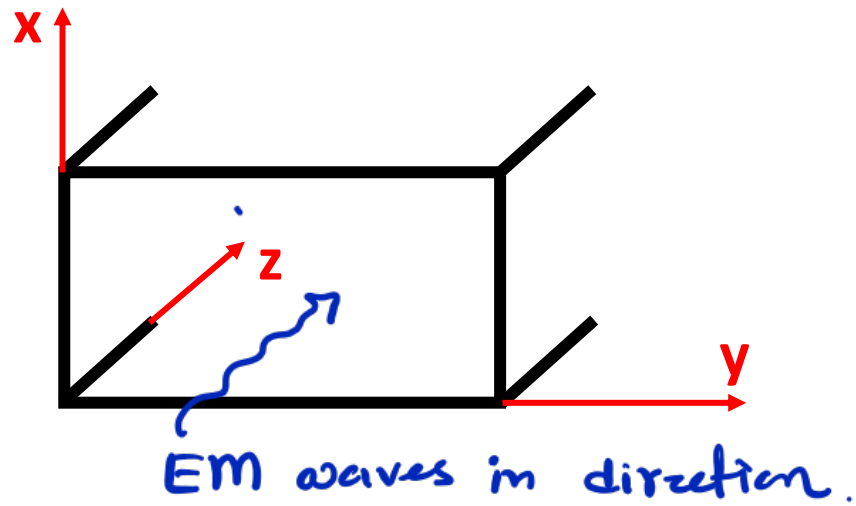


Single Ridged Waveguide



Double Ridged Waveguide

Why TEM Mode Can not Propagated in Rectangular Waveguides



→ In TEM mode, If direction of propagation is z

$$\Rightarrow E_z = H_z = 0$$

$$\Rightarrow E_x, E_y, H_x \text{ \& } H_y \neq 0$$

→ For TE mode wave eq.ⁿ

$$\Rightarrow \nabla^2 H_z = -\omega^2 \mu \epsilon H_z \text{ — (A)}$$

→ For TM mode wave eq.ⁿ

$$\Rightarrow \nabla^2 E_z = -\omega^2 \mu \epsilon E_z \text{ — (B)}$$

→ From eqⁿ (1)

$$\Rightarrow \frac{\partial^2 H_2}{\partial x^2} + \frac{\partial^2 H_2}{\partial y^2} + \frac{\partial^2 H_2}{\partial z^2} = -\omega^2 \mu \epsilon H_2$$

$$[\text{Hence, } \frac{\partial^2}{\partial z^2} = \gamma^2]$$

$$\Rightarrow \frac{\partial^2 H_2}{\partial x^2} + \frac{\partial^2 H_2}{\partial y^2} + \gamma^2 H_2 + \omega^2 \mu \epsilon H_2 = 0$$

$$\Rightarrow \frac{\partial^2 H_2}{\partial x^2} + \frac{\partial^2 H_2}{\partial y^2} + (\gamma^2 + \omega^2 \mu \epsilon) H_2 = 0$$

$$[\text{Consider, } \gamma^2 + \omega^2 \mu \epsilon = h^2]$$

$$\Rightarrow \frac{\partial^2 H_2}{\partial x^2} + \frac{\partial^2 H_2}{\partial y^2} + h^2 H_2 = 0 \quad \text{--- (1)}$$

→ Similarly,

$$\Rightarrow \frac{\partial^2 E_2}{\partial x^2} + \frac{\partial^2 E_2}{\partial y^2} + h^2 E_2 = 0 \quad \text{--- (2)}$$

→ From Maxwell's eq.ⁿ

$$\Rightarrow \nabla \times \vec{H} = j\omega \epsilon \vec{E}$$

$$\Rightarrow \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ H_x & H_y & H_z \end{vmatrix} = j\omega \epsilon (E_x \hat{i} + E_y \hat{j} + E_z \hat{k})$$

$$[\text{Hence, } \frac{\partial}{\partial z} = -\sigma]$$

$$\Rightarrow \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & -\sigma \\ H_x & H_y & H_z \end{vmatrix} = j\omega \epsilon (E_x \hat{i} + E_y \hat{j} + E_z \hat{k})$$

→ $\hat{i}$ Component

$$\Rightarrow \frac{\partial H_z}{\partial y} + \sigma H_y = j\omega \epsilon E_x$$

— (3)

→ $\hat{j}$ Component

$$\Rightarrow \frac{\partial H_z}{\partial x} + \sigma H_x = -j\omega \epsilon E_y$$

— (4)

→ $\hat{k}$ Component

$$\Rightarrow \frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y} = j\omega \epsilon E_z$$

— (5)

→ As per Maxwell's eqⁿ

$$\Rightarrow \nabla \times \vec{E} = -j\omega\mu\vec{H}$$

$$\Rightarrow \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ E_x & E_y & E_z \end{vmatrix} = -j\omega\mu [H_x \hat{i} + H_y \hat{j} + H_z \hat{k}]$$

$$[H_{enc}, \frac{\partial}{\partial z} = -\sigma]$$

$$\Rightarrow \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & -\sigma \\ E_x & E_y & E_z \end{vmatrix} = -j\omega\mu [H_x \hat{i} + H_y \hat{j} + H_z \hat{k}]$$

→ $\hat{i}$ component

$$\Rightarrow \frac{\partial E_z}{\partial y} + \sigma E_y = -j\omega\mu H_x$$

— (6)

→ $\hat{j}$ component

$$\Rightarrow \frac{\partial E_z}{\partial x} + \sigma E_x = j\omega\mu H_y$$

— (7)

→ $\hat{k}$ component

$$\Rightarrow \frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} = -j\omega\mu H_z$$

— (8)

∴ Sol.ⁿ of (1), (2), (3), (4), (5), (6), (7) & (8) is given by

$$\Rightarrow E_x = -\frac{\gamma}{h^2} \frac{\partial E_2}{\partial x} - j \frac{\omega \mu}{h^2} \frac{\partial H_2}{\partial y}$$

$$\Rightarrow E_y = -\frac{\gamma}{h^2} \frac{\partial E_2}{\partial y} + j \frac{\omega \mu}{h^2} \frac{\partial H_2}{\partial x}$$

$$\Rightarrow H_x = -\frac{\gamma}{h^2} \frac{\partial H_2}{\partial x} + j \frac{\omega \epsilon}{h^2} \frac{\partial E_2}{\partial y}$$

$$\Rightarrow H_y = -\frac{\gamma}{h^2} \frac{\partial H_2}{\partial y} - j \frac{\omega \epsilon}{h^2} \frac{\partial E_2}{\partial x}$$

∴ With TEM mode, $E_2 = H_2 = 0$

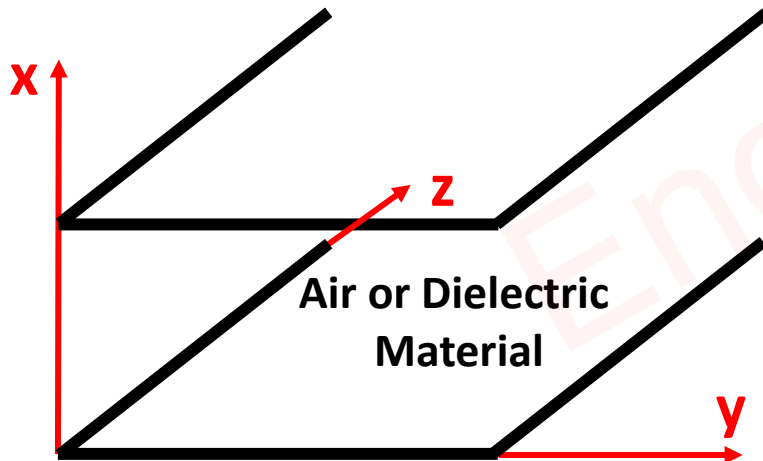
$$\text{Hence, } E_x = H_x = E_y = H_y = 0$$

∴ Hence, TEM mode is not possible in rectangular waveguide.

Parallel Plate Waveguide

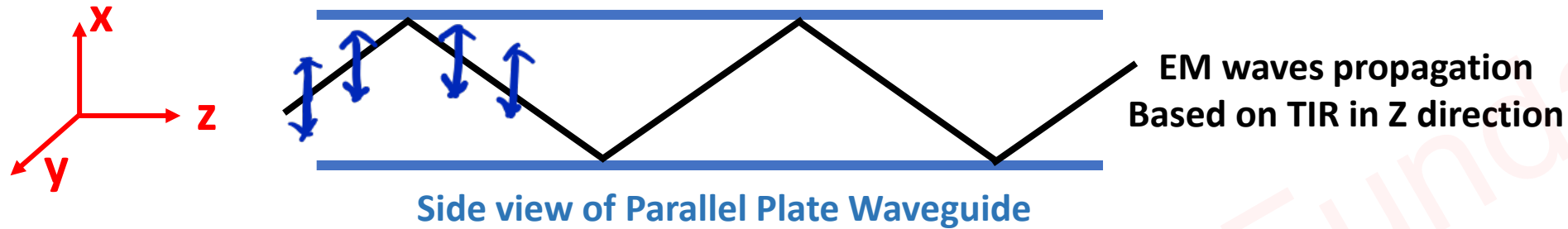
Basics of Parallel Plate Waveguide

- ❑ Parallel Plate Waveguide has Two Metal Plates in Parallel separated by Air or Dielectric Material.
- ❑ In Parallel Plate Waveguide, EM Waves propagate due to Total Internal Reflection (TIR).
- ❑ In Parallel Plate Waveguide, Fields are contained between Two Parallel Plates.
- ❑ In Parallel Plate Waveguide, TE TM and TEM Modes can propagate.
- ❑ It offers high bandwidth with a single Mode.



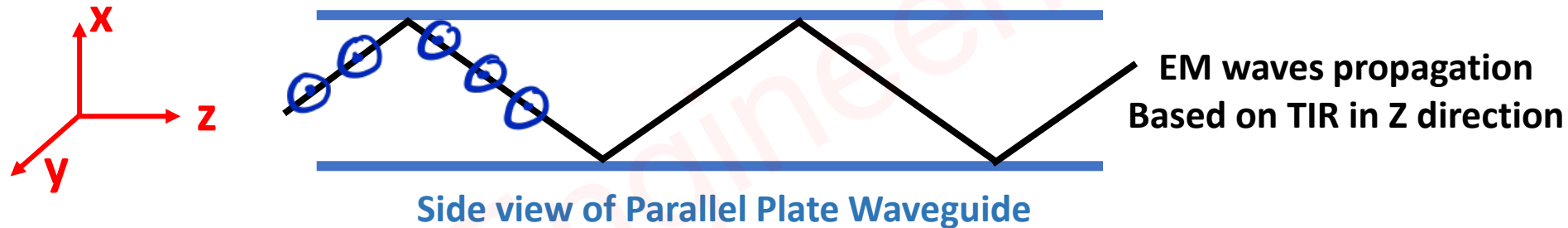
Modes of Parallel Plate Waveguide

□ TE Mode in Parallel Plate Waveguide



$E_x(z, t) \perp$ direction of propagation z

□ TM Mode in Parallel Plate Waveguide

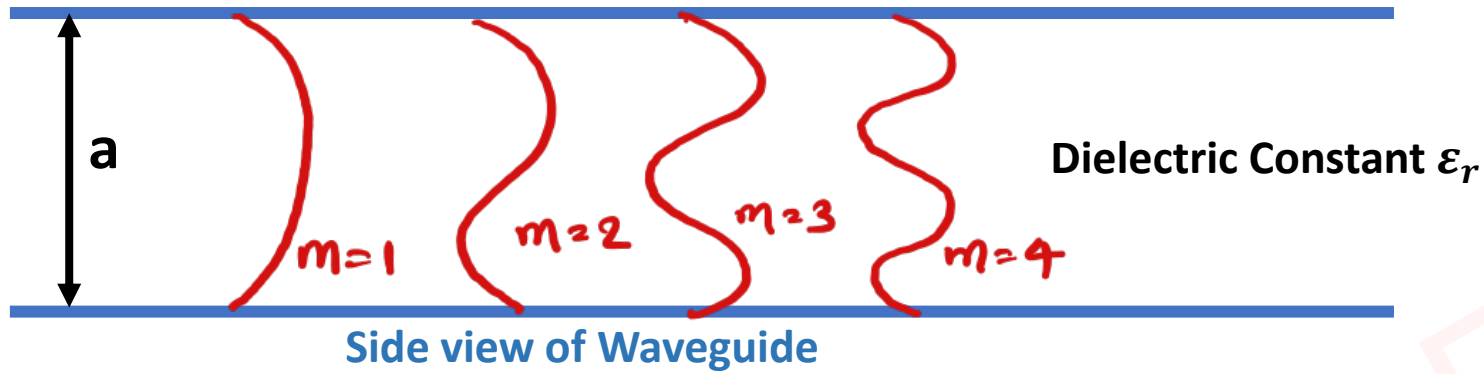


$H_y(z, t) \perp$ direction of propagation z

□ TEM Mode can also propagate in Parallel Plate Waveguide

$E_x(z, t), H_y(z, t) \perp$ direction of propagation z .

Cut-Off Frequency of Parallel Plate Waveguide



$$\Rightarrow f_c = \frac{mc}{2a\sqrt{\epsilon_r}}, \quad m = 1, 2, 3, \dots$$

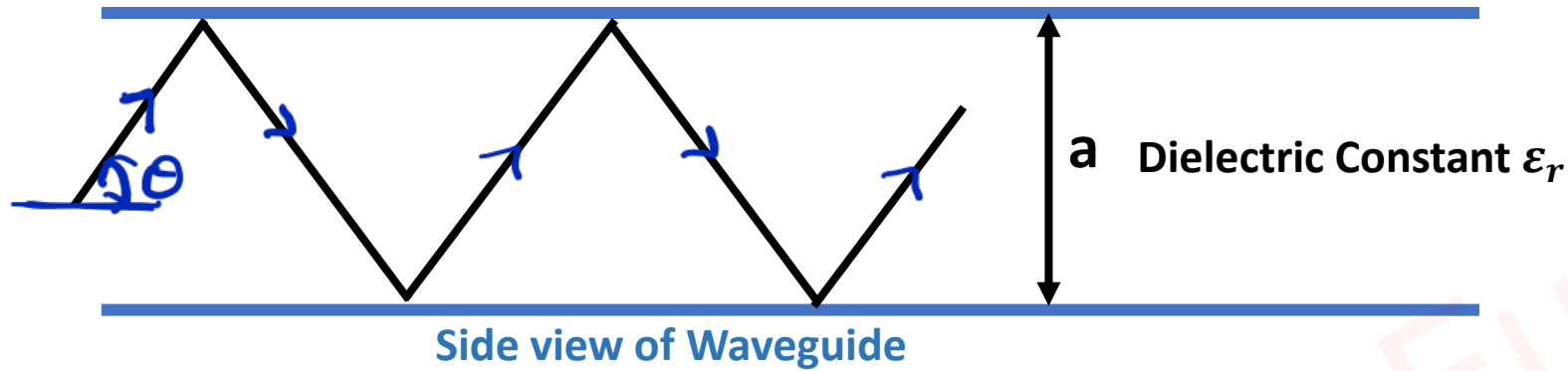
where, m = Operating mode

c = 3×10^8 m/sec

a = Spacing betⁿ two plates

ϵ_r = Dielectric Constant.

Tilt Angle of Parallel Plate Waveguide



EM waves propagation
Based on TIR in Z direction

□ It is the Angle of incidence of EM wave with respect to the direction of propagation.

$$\Rightarrow \sin \theta = \frac{f_c}{f}$$

where, θ = Tilt angle

f_c = Cut off freq.

f = Operating freq.

Intrinsic Wave Impedance of Parallel Plate Waveguide

□ Intrinsic Wave Impedance of TE Mode η_{TE}

$$\Rightarrow \eta_{TE} = \frac{120\pi}{\sqrt{1 - \left(\frac{f_c}{f}\right)^2}} = \frac{120\pi}{\sqrt{1 - \sin^2\theta}} = \frac{120\pi}{\cos\theta}$$

□ Intrinsic Wave Impedance of TM Mode η_{TM}

$$\Rightarrow \eta_{TM} = 120\pi \sqrt{1 - \left(\frac{f_c}{f}\right)^2} = 120\pi \sqrt{1 - \sin^2\theta} = 120\pi \cos\theta$$

□ Intrinsic Wave Impedance of TEM Mode η_{TEM}

$$\Rightarrow \eta_{TEM} = 120\pi$$

Phase Velocity and Group Velocity of Parallel Plate Waveguide

□ Phase Velocity of Parallel Plate Waveguide v_p

$$\leadsto v_p = \frac{c}{\sqrt{1 - \left(\frac{f_c}{f}\right)^2}}$$

□ Group Velocity of Parallel Plate Waveguide v_g

$$\leadsto v_g = c \sqrt{1 - \left(\frac{f_c}{f}\right)^2}$$

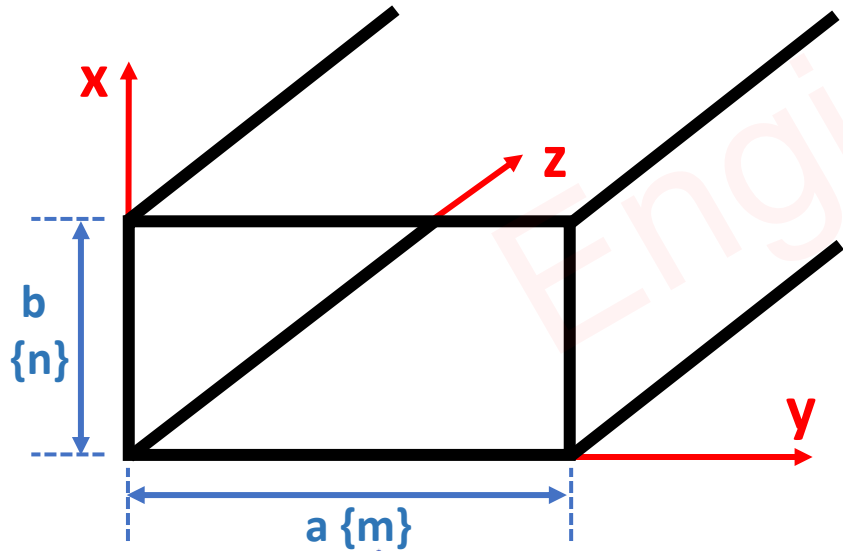
□ Relation between Phase Velocity and Group Velocity of Parallel Plate Waveguide:

$$\leadsto v_p v_g = c^2$$

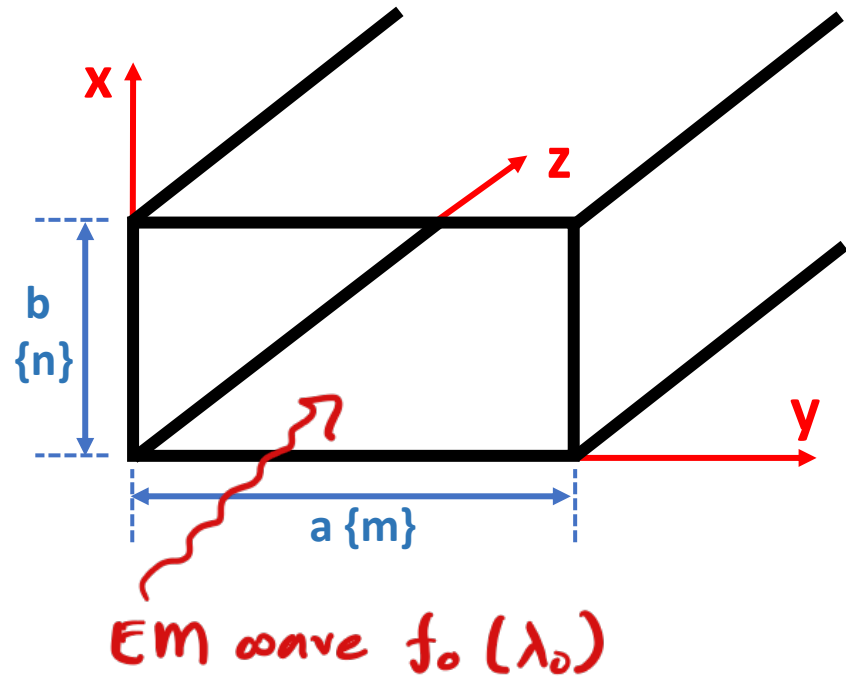
Rectangular Waveguide

Basics of Rectangular Waveguide

- ❑ Waveguide is a Hollow Metallic tube with a uniform cross-section.
- ❑ Waveguide is used to transmit EM wave using successive reflections from inner walls of waveguide.
- ❑ Rectangular Waveguide has less Attenuation Loss compared to Transmission Line.
- ❑ Rectangular Waveguide allows only TE_{mn} and TM_{mn} Modes, It does not allow TEM mode.
- ❑ Dominant Mode in Rectangular Waveguide is TE_{10} , where, $m=1$ and $n=0$.



Cut-off Frequency and Cut-off Wavelength of Rectangular Waveguide



$$\approx f_c = \frac{c}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$$

$$\approx \lambda_c = \frac{2ab}{\sqrt{m^2 b^2 + n^2 a^2}}$$

$$\approx \text{For TE}_{10}, m=1 \text{ \& } n=0$$

$$\approx f_{c\text{TE}_{10}} = \frac{c}{2a}, \quad \lambda_{c\text{TE}_{10}} = 2a$$

□ Condition for wave propagation:

$$\approx \lambda_c > \lambda_0$$

↑ ↑
(cut-off Operating
wavelength wavelength)

$$\approx f_c < f_0$$

↑ ↑
(cut off Operating
freq. freq.)

Checking of Modes in Rectangular Waveguide

→ Lowest cut-off freq.

or

Highest cut-off wavelength.

TE_{10}	TE_{01}	TE_{11}	TE_{21}	TE_{12}	TE_{22}	
TM_{10}	TM_{01}	TM_{11}	TM_{21}	TM_{12}	TM_{22}	

X

→ It is not possible.

→ These are even Scent modes.

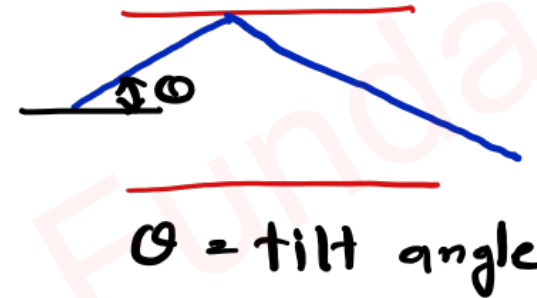
Check for
These modes
atleast.

Intrinsic Wave Impedance of Rectangular Waveguide

□ Intrinsic Wave Impedance of TE Mode η_{TE}

$$\leadsto \eta_{TE} = \frac{120\pi}{\sqrt{1 - \left(\frac{f_c}{f}\right)^2}} = \frac{120\pi}{\sqrt{1 - \sin^2 \theta}} = \frac{120\pi}{\cos \theta}$$

$$[\text{Here, } \frac{f_c}{f} = \sin \theta]$$



□ Intrinsic Wave Impedance of TM Mode η_{TM}

$$\leadsto \eta_{TM} = 120\pi \sqrt{1 - \left(\frac{f_c}{f}\right)^2} = 120\pi \cos \theta$$

$$[\text{Here, } \frac{f_c}{f} = \sin \theta]$$

Phase Velocity and Group Velocity of Rectangular Waveguide

□ Phase Velocity of Rectangular Waveguide v_p

$$\Rightarrow v_p = \frac{c}{\sqrt{1 - \left(\frac{f_c}{f}\right)^2}} = \frac{c}{\sqrt{1 - \left(\frac{\lambda_0}{\lambda_c}\right)^2}}$$

□ Group Velocity of Rectangular Waveguide v_g

$$\Rightarrow v_g = c \sqrt{1 - \left(\frac{f_c}{f}\right)^2} = c \sqrt{1 - \left(\frac{\lambda_0}{\lambda_c}\right)^2}$$

□ Relation between Phase Velocity and Group Velocity of Rectangular Waveguide:

$$\Rightarrow v_p v_g = c^2$$

□ Guided Wavelength of Rectangular Waveguide:

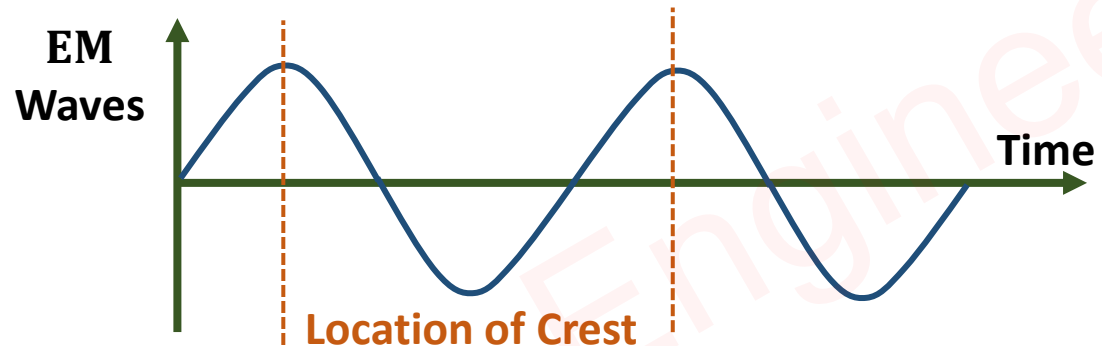
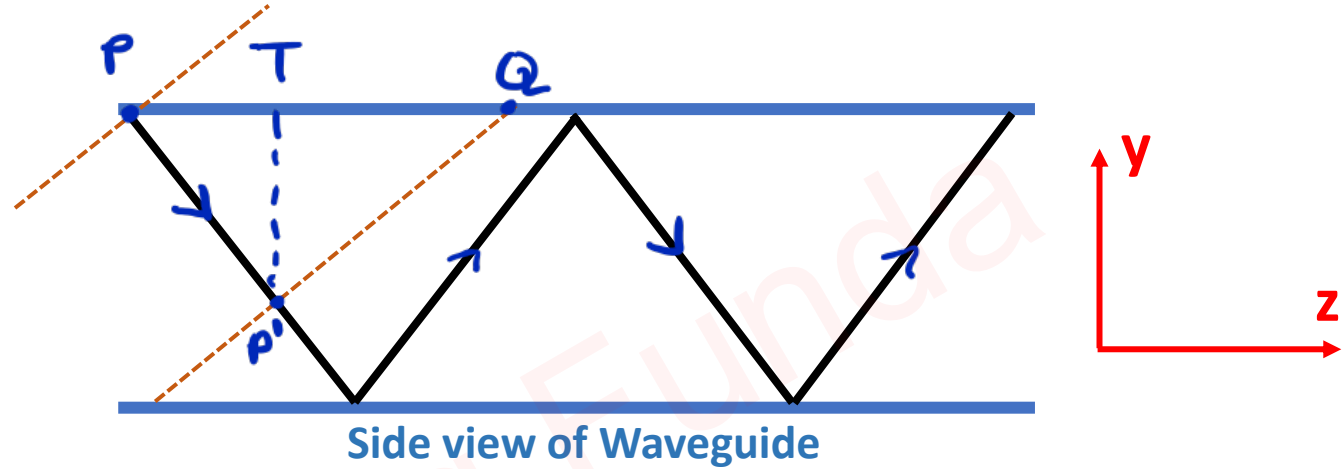
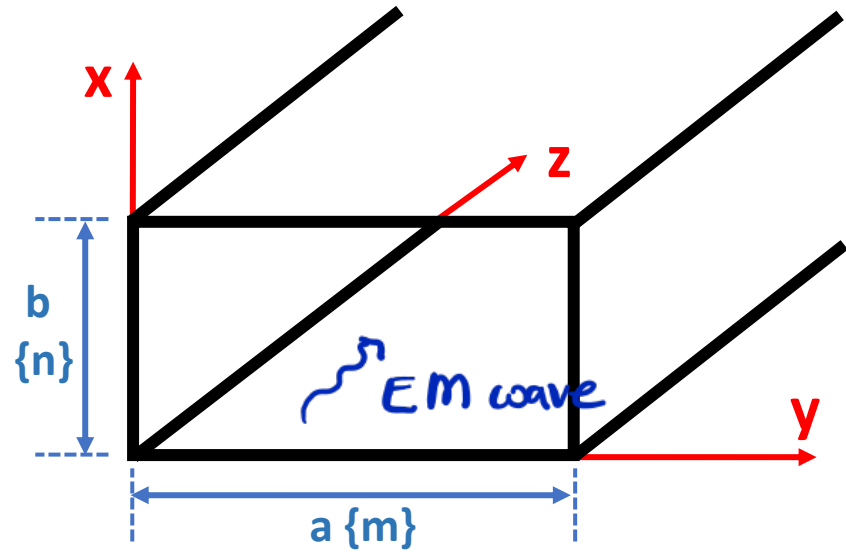
$$\Rightarrow \lambda_g = \frac{\lambda_0}{\sqrt{1 - \left(\frac{\lambda_0}{\lambda_c}\right)^2}}$$

Group Velocity & Phase Velocity

Analogy of Group Velocity and Phase Velocity

- ❑ Let us consider the City Marathon runners as individual EM waves.
- ❑ As Time Passes in the race, Runners spread out (Disperse) simply because each runner in the group is running at a different speed.
- ❑ The phase velocity is like the velocity of an individual runner and The group velocity is like the velocity of a group of runners.
- ❑ Most often, individual runners can run faster than the group as a whole, This means V_p will be higher than V_g .
- ❑ However, this depends on the properties of the medium.
 - If $V_p > V_g$, Medium is a dispersive medium.
 - If $V_p = V_g$, Medium is a non-dispersive medium.
 - If $V_p < V_g$, Medium is an anomalous dispersive medium.

Physical meaning of Group Velocity and Phase Velocity



$$v_p = \frac{d\phi}{dt} > c$$
$$v_g = \frac{d\phi}{dt} < c$$

❑ The Phase velocity is associated with the phase of EM wave (Guided Wavelength) and The Group velocity is associated with the group of EM wave propagation.

Calculation of Phase Velocity and Group Velocity

□ Phase Velocity of Rectangular Waveguide v_p

$$\rightarrow v_p = \frac{\omega}{\beta} = \frac{2\pi f}{2\pi/\lambda_g} = \frac{\lambda_0 f}{\sqrt{1 - \left(\frac{\lambda_0}{\lambda_c}\right)^2}} = \frac{c}{\sqrt{1 - \left(\frac{\lambda_0}{\lambda_c}\right)^2}} = \frac{c}{\sqrt{1 - \left(\frac{f_c}{f}\right)^2}}$$

$\left[\text{Here, } \lambda_g = \frac{\lambda_0}{\sqrt{1 - \left(\frac{\lambda_0}{\lambda_c}\right)^2}} \right] \quad [f > f_c]$

□ Group Velocity of Rectangular Waveguide v_g

$$\rightarrow v_g = \frac{d\omega}{d\beta}$$
$$= c \sqrt{1 - \left(\frac{\lambda_0}{\lambda_c}\right)^2} = c \sqrt{1 - \left(\frac{f_c}{f}\right)^2}$$
$$\rightarrow v_p v_g = c^2$$

Relation of Phase Velocity and Group Velocity

$$\leadsto v_g = \frac{d\omega}{d\beta} = \frac{d(\omega/\beta)\beta}{d\beta} = \frac{d(v_p \beta)}{d\beta}$$

$$\left[\frac{d(AB)}{dt} = A \frac{dB}{dt} + B \frac{dA}{dt} \right]$$

$$\leadsto v_g = v_p + \beta \frac{dv_p}{d\beta}$$

$$\left[\beta = \frac{2\pi}{\lambda_g} \right]$$

$$\begin{aligned} \leadsto v_g &= v_p + \left(\frac{2\pi}{\lambda_g} \right) \frac{dv_p}{d\left(\frac{2\pi}{\lambda_g} \right)} \\ &= v_p + \left(\frac{\cancel{2\pi}}{\cancel{\lambda_g}} \right) \frac{dv_p}{(\cancel{2\pi}) \left(\frac{-1}{\lambda_g^2} \right) d\lambda_g} \end{aligned}$$

$$\leadsto \boxed{v_g = v_p - \lambda_g \frac{dv_p}{d\lambda_g}}$$

Examples of Group velocity & Phase Velocity

Question 1 – The dispersion of a certain wave is given by:

$$\omega = \sqrt{c^2 k^2 + m^2}$$

Where ω is the angular frequency, k is the wave vector, c is the velocity of light and m is a constant. The group velocity v of the wave has the following properties:

- a. $v \rightarrow c$ as $k \rightarrow 0$ and $v \rightarrow c$ as $k \rightarrow \infty$
- b. $v \rightarrow c$ as $k \rightarrow 0$ and $v \rightarrow \infty$ as $k \rightarrow \infty$
- c. $v \rightarrow 0$ as $k \rightarrow 0$ and $v \rightarrow \infty$ as $k \rightarrow \infty$
- ✓ d. $v \rightarrow 0$ as $k \rightarrow 0$ and $v \rightarrow c$ as $k \rightarrow \infty$

$$\begin{aligned} \Rightarrow v &= \frac{d\omega}{dk} = \frac{d\sqrt{c^2 k^2 + m^2}}{dk} \\ &= \left(\frac{1}{2}\right) (c^2 k^2 + m^2)^{\frac{1}{2}-1} (c^2 (2k)) \\ &= \frac{c^2 k}{\sqrt{c^2 k^2 + m^2}} = \frac{c^2}{\sqrt{c^2 + (m/k)^2}} \end{aligned}$$

$$\leadsto k \rightarrow 0$$

$$\Rightarrow \theta = \lim_{k \rightarrow 0} \frac{c^2 k}{\sqrt{c^2 k^2 + m^2}}$$

$$= \frac{0}{\sqrt{0 + m^2}} = 0$$

$$\boxed{\leadsto k \rightarrow 0, \theta \rightarrow 0}$$

$$\leadsto k \rightarrow \infty$$

$$\Rightarrow \theta = \lim_{k \rightarrow \infty} \frac{c^2}{\sqrt{c^2 + (v/k)^2}}$$

$$= \frac{c^2}{\sqrt{c^2 + 0}} = c$$

$$\boxed{\leadsto k \rightarrow \infty, \theta \rightarrow c}$$

Question 2 – Given a wave with the dispersion relation $\omega = ck + m$ for $k > 0$ and $m > 0$, which one of the following is true?

- a. The group velocity is greater than the phase velocity
- ✓ b. The group velocity is less than the phase velocity
- c. The group velocity is equal to phase velocity
- d. No relation in between group velocity and phase velocity

$$\leadsto v_g = \frac{d\omega}{dk} = \frac{d(ck+m)}{dk}$$

$$= c + 0$$

$$\boxed{v_g = c}$$

$$\boxed{\leadsto v_p > v_g}$$

$$\leadsto v_p = \frac{\omega}{k} = \frac{ck+m}{k}$$

$$\boxed{v_p = c + \frac{m}{k}}$$

Question 3 – The angular frequency ω and wavelength relation is given by:

$$\omega \propto 1/\sqrt{\lambda}$$

Which of the relations between group velocity v_g and phase velocity v_p is true?

☒ a. $v_g = v_p/2$

☐ b. $v_g = v_p$

☐ c. $v_g = 2v_p$

☐ d. No relation exists

$$\leadsto \omega \propto \frac{1}{\sqrt{\lambda}}$$

$$\leadsto k = \frac{2\pi}{\lambda} \Rightarrow \lambda = \frac{2\pi}{k}$$

$$\leadsto \omega \propto \sqrt{k}$$

$$\Rightarrow \omega = A\sqrt{k}$$

$$\leadsto v_p = \frac{\omega}{k} = \frac{A\sqrt{k}}{k} = \frac{A}{\sqrt{k}}$$

$$\begin{aligned} \leadsto v_g &= \frac{d\omega}{dk} = \frac{d(A\sqrt{k})}{dk} = A\left(\frac{1}{2}\right)k^{-1/2} \\ &= \frac{A}{2\sqrt{k}} = \frac{v_p}{2} \end{aligned}$$

Question 4 – Phase velocity and group for a wave will be identical:

- a. ✓ If the phase velocity is independent of wavelength
- b. If the phase velocity is dependent of wavelength
- c. If the group velocity is independent of wavelength
- d. If the group velocity is dependent of wavelength

$$\leadsto v_p = v_g$$

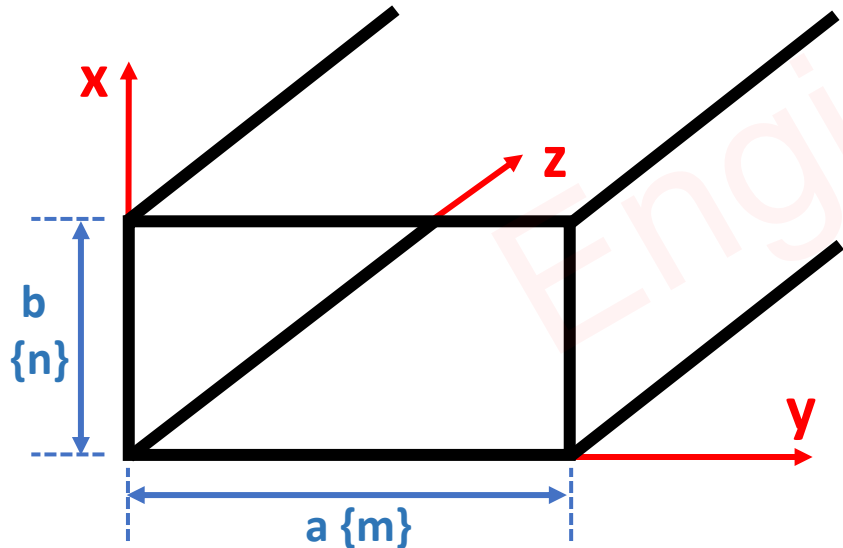
$$\rightarrow v_g = v_p - \lambda \frac{dv_p}{d\lambda} \Rightarrow v_g = v_p$$

$$[\text{If } v_p \text{ is independent on } \lambda \text{ then } \frac{dv_p}{d\lambda} = 0]$$

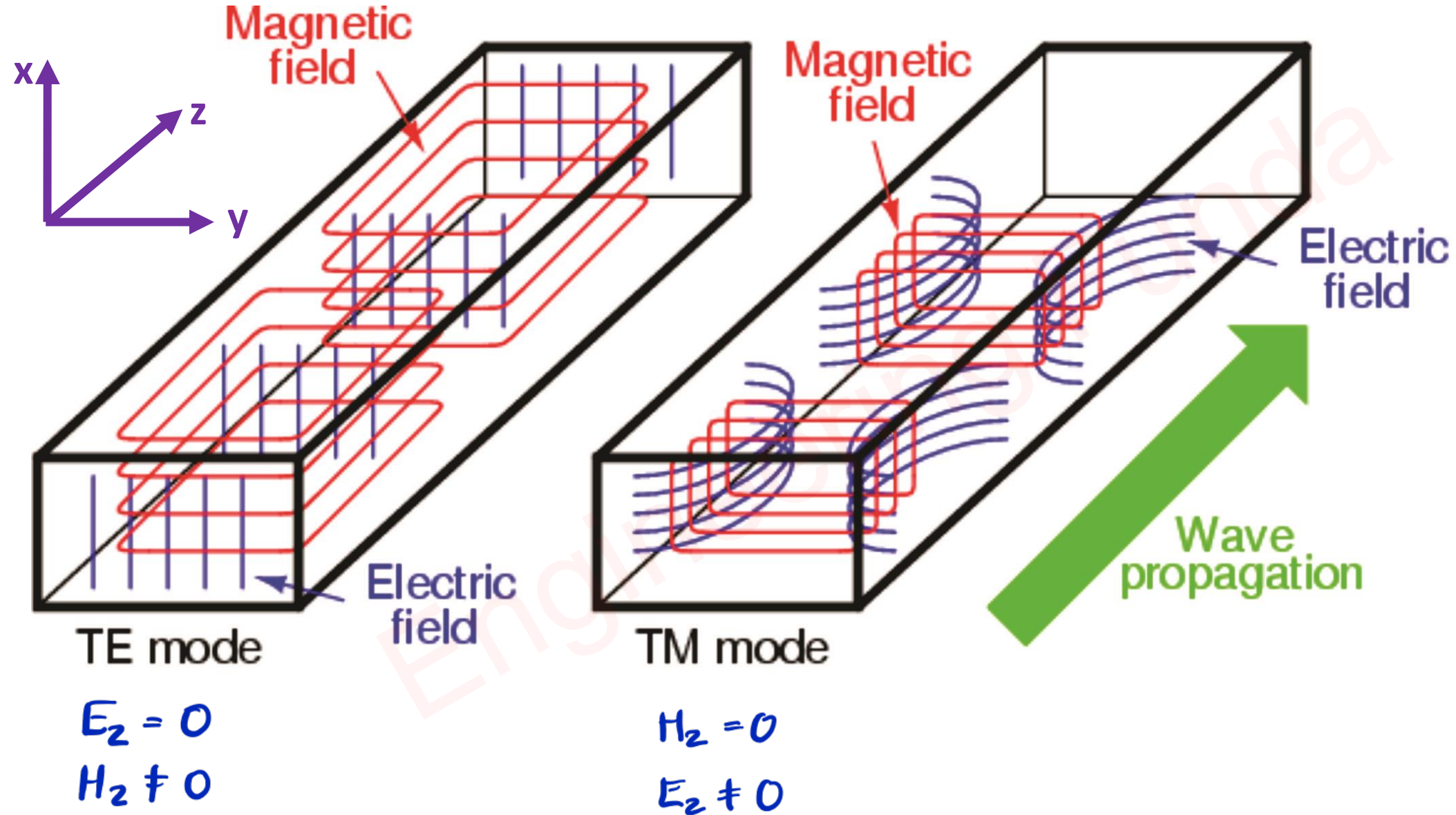
Modes in Rectangular Waveguide

Basics of Rectangular Waveguide

- ❑ Waveguide is a Hollow Metallic tube with a uniform cross-section.
- ❑ Waveguide is used to transmit EM wave using successive reflections from inner walls of waveguide.
- ❑ Rectangular Waveguide allows TE_{mn} and TM_{mn} Modes, It does not allow TEM mode.
- ❑ **Dominant Mode** in Rectangular Waveguide is TE_{10} , where, $m=1$ and $n=0$.
- ❑ m and n defines order of Mode in Rectangular waveguide.



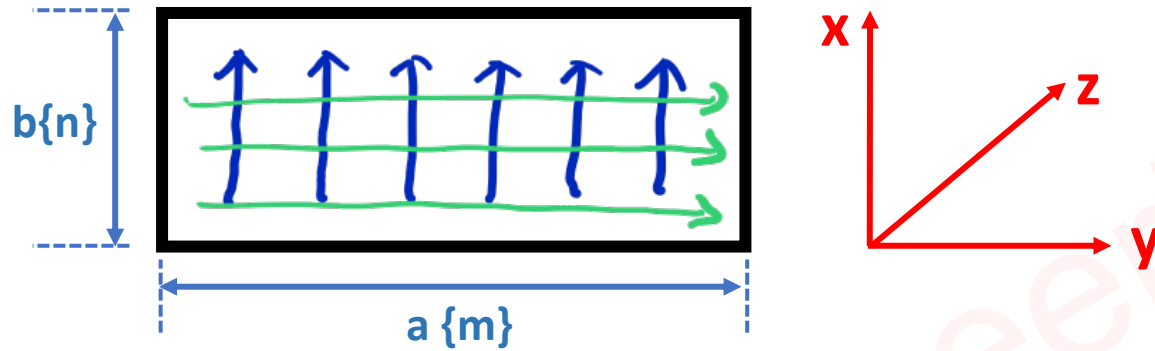
TE and TM mode of Rectangular Waveguide



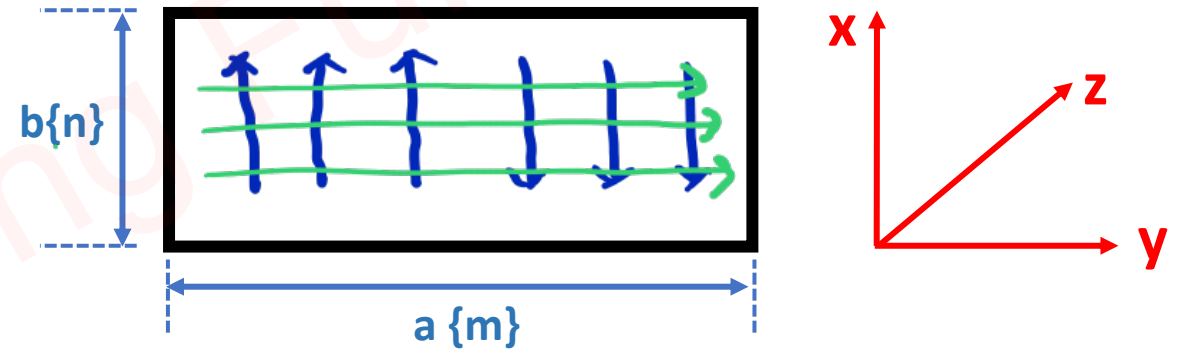
TE mode of Rectangular Waveguide

- TE mode is Transverse Electric Mode.
- TE mode is based on **Voltage Wave**.
- In TE mode, for the direction of propagation in the z direction, $E_z = 0$ and $H_z \neq 0$

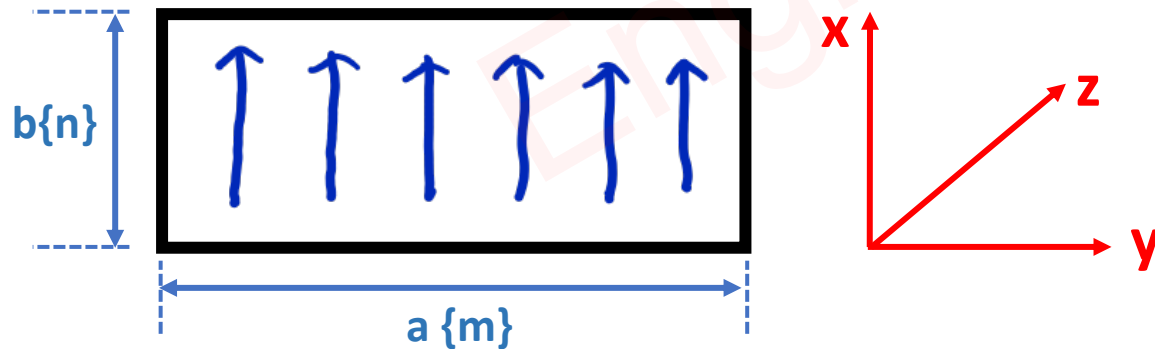
TE₁₁ $m=1, n=1$



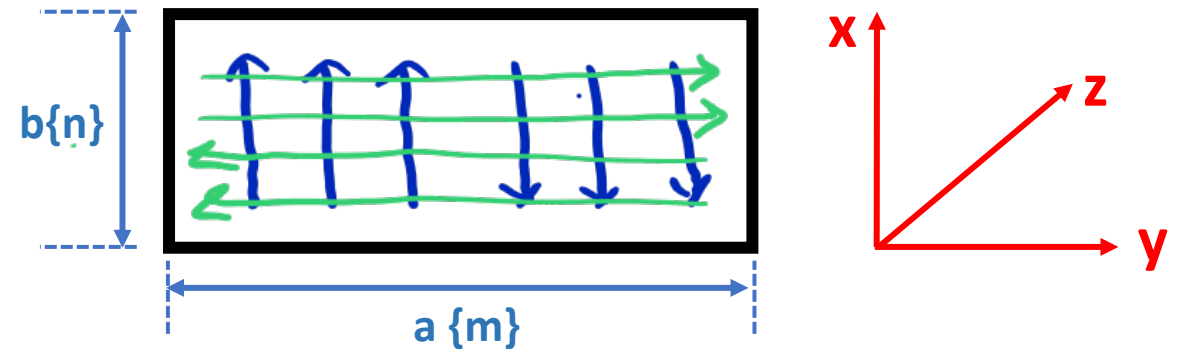
TE₂₁ $m=2, n=1$



TE₁₀ ($m=1, n=0$)

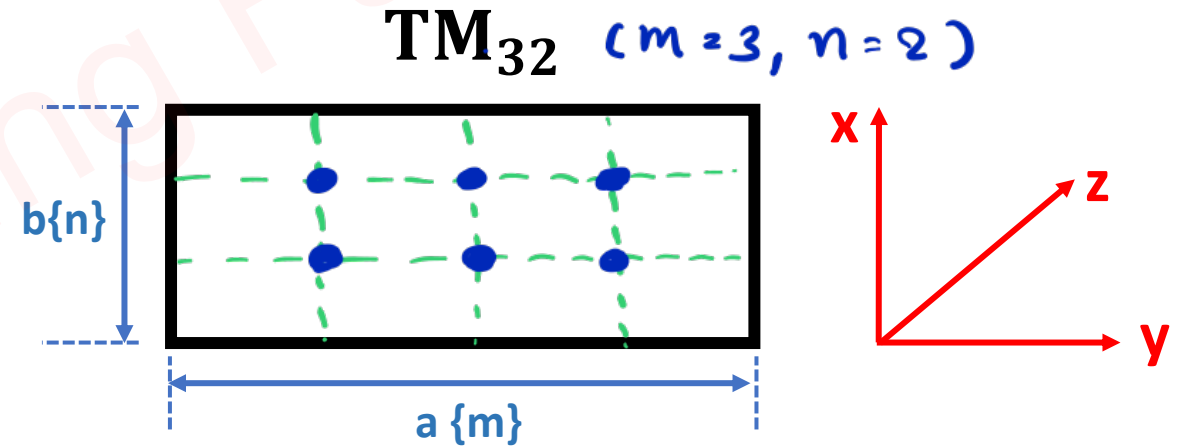
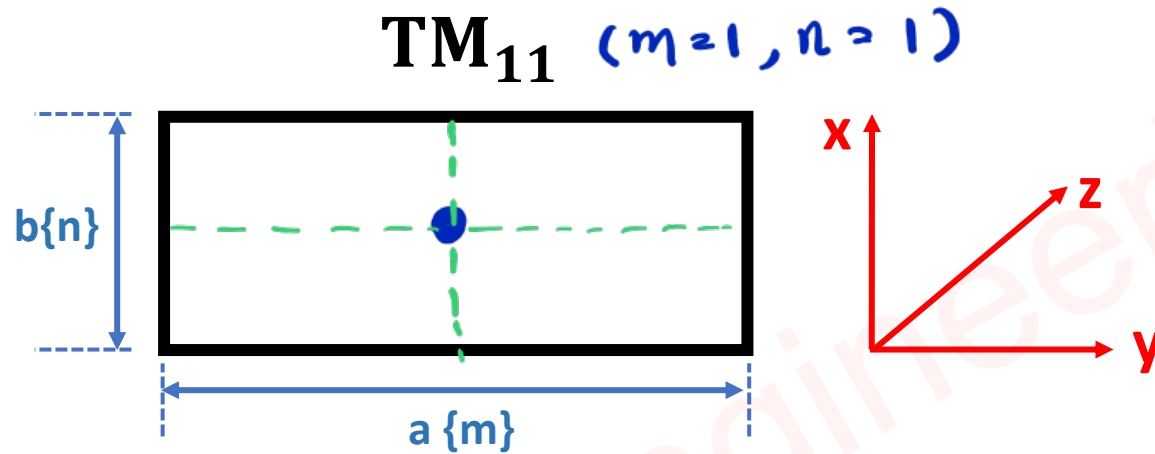


TE₂₂ ($m=2, n=2$)



TM mode of Rectangular Waveguide

- ❑ TM mode is Transverse Magnetic Mode.
- ❑ TM mode is based on the **Current Wave**.
- ❑ In TM mode, for the direction of propagation in the z direction, $H_z = 0$ and $E_z \neq 0$
- ❑ In rectangular waveguide, TM_{10} and TM_{01} modes are not possible, these are **even scant modes**.

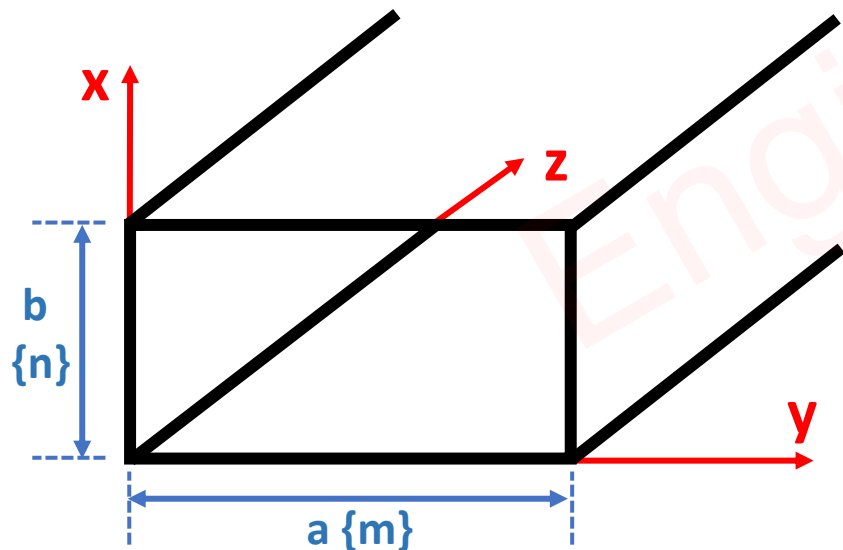


Degenerated modes of Rectangular Waveguide

- ❑ If two or modes have the same cut of frequencies then they are Degenerated modes of rectangular waveguide.
- ❑ Cut-off frequency of rectangular waveguide can be calculated by:

$$f_c = \frac{c}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$$

- ❑ Here, m and n define the order of mode.
- ❑ Only TE and TM modes can propagate through rectangular waveguides, TEM mode is not possible in rectangular waveguides.



$$\leadsto TE_{11} \leftrightarrow TM_{11}$$

These are degenerated modes.

$$\leadsto a = b$$

$$TE_{23} \leftrightarrow TE_{32}$$

These are degenerated modes.

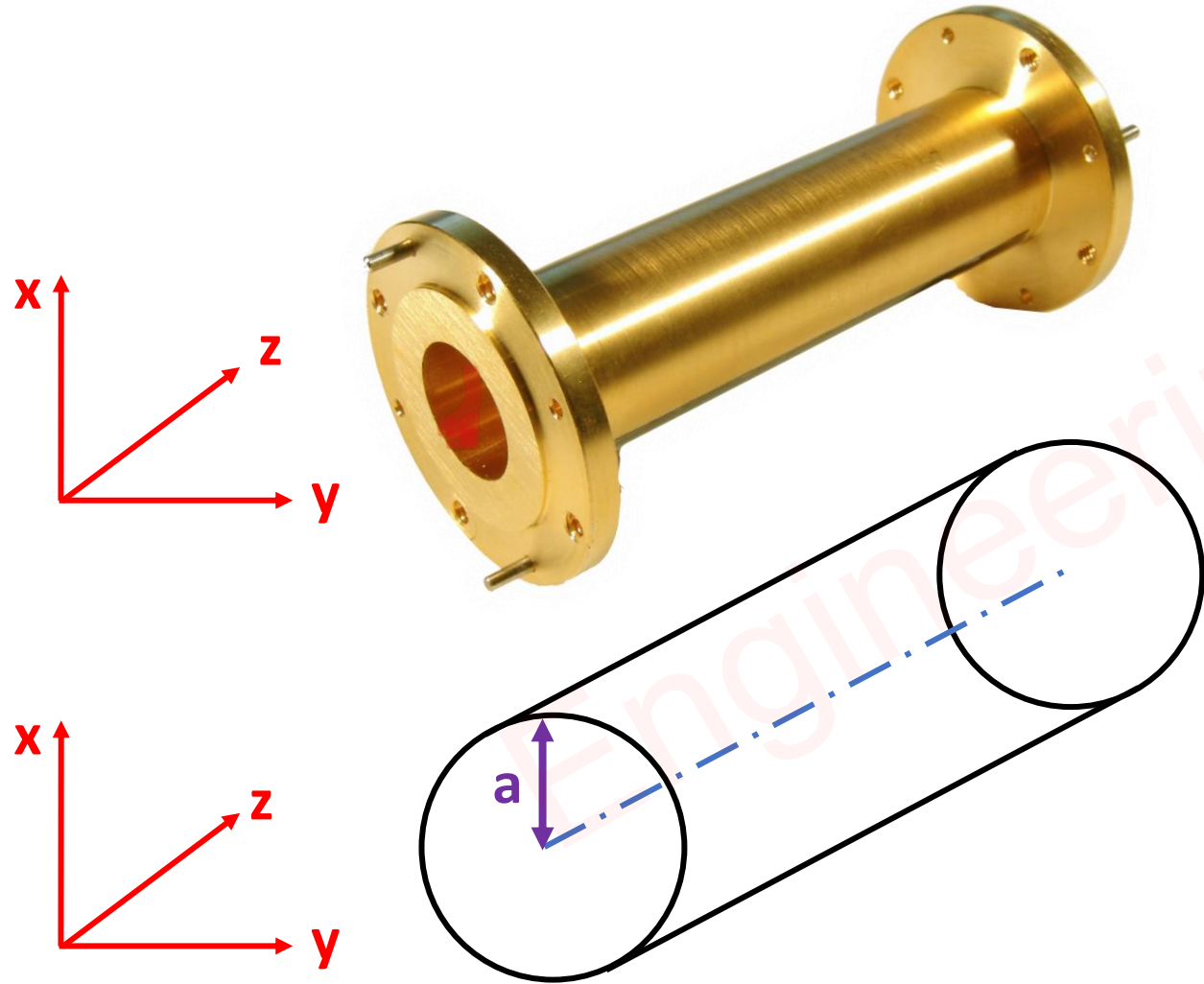
Circular Waveguide

Basics of Circular Waveguide

- ❑ Circular Waveguide is easier to manufacture compared to a rectangular waveguide.
- ❑ Circular Waveguide can be easily joined compared to a rectangular waveguide.
- ❑ Circular Waveguides occupy more space compared to rectangular waveguides.
- ❑ Dominant Mode of Circular Waveguide is TE_{11} .
- ❑ It has more Interference at dominant mode compared to rectangular waveguides.
- ❑ In Circular Waveguide, TE_{01} and TM_{01} modes are rotationally symmetrical. Hence rotation of polarization can be overcome. However, All modes are not rotationally symmetrical.
- ❑ In Circular Waveguide, TE_{01} has lowest attenuation. So, it is suitable for long waveguide transmission.
- ❑ Circular Waveguide allows TE_{mn} and TM_{mn} Modes, It does not allow TEM mode.

Structure of Circular Waveguide

- ❑ Circular Waveguide is a hollow metallic tube with a circular uniform cross-section.
- ❑ Here, in the figure, the radius of the circular waveguide is a .



Designing Parameters of Circular Waveguide

□ Cut-Off Frequency of Circular Waveguide

$$f_c = \frac{\chi_{np}}{2\pi a \sqrt{\mu\epsilon}} = \frac{\chi_{np}}{2\pi a} \times v \quad \leadsto v = \frac{1}{\sqrt{\mu\epsilon}} = \frac{1}{\sqrt{\mu_0 \mu_r \epsilon_0 \epsilon_r}} = \frac{c}{\sqrt{\mu_r \epsilon_r}}$$

- χ_{np} is based on Bessel's equation {For TE_{11} Mode, $\chi_{np} = 1.841$ }
- a is radius of Circular waveguide
- v is velocity of EM wave

□ Cut-Off Wavelength of Circular Waveguide

$$\lambda_c = \frac{v}{f_c} = \frac{2\pi a}{\chi_{np}}$$

□ Phase Velocity of Circular Waveguide

$$v_p = \frac{v}{\sqrt{1 - \left(\frac{f_c}{f}\right)^2}}$$

□ Guided Wavelength of Circular Waveguide

$$\lambda_g = \frac{\lambda}{\sqrt{1 - \left(\frac{f_c}{f}\right)^2}}$$

□ Wave Impedance of Circular Waveguide

$$(Z_w)_{TE} = \eta \sqrt{1 - \left(\frac{f_c}{f}\right)^2} \quad \leadsto (Z_w)_{TE} = \frac{\eta}{\sqrt{1 - (f_c/f)^2}}$$

- η is free space impedance and it's value is 120π .

Applications of Circular Waveguide

- ❑ To connect horn Antenna for feeding a Reflector Antenna.
- ❑ Short and Medium distance broad band communication.
- ❑ TE_{01} is used for long distance waveguide transmission above 10GHz.

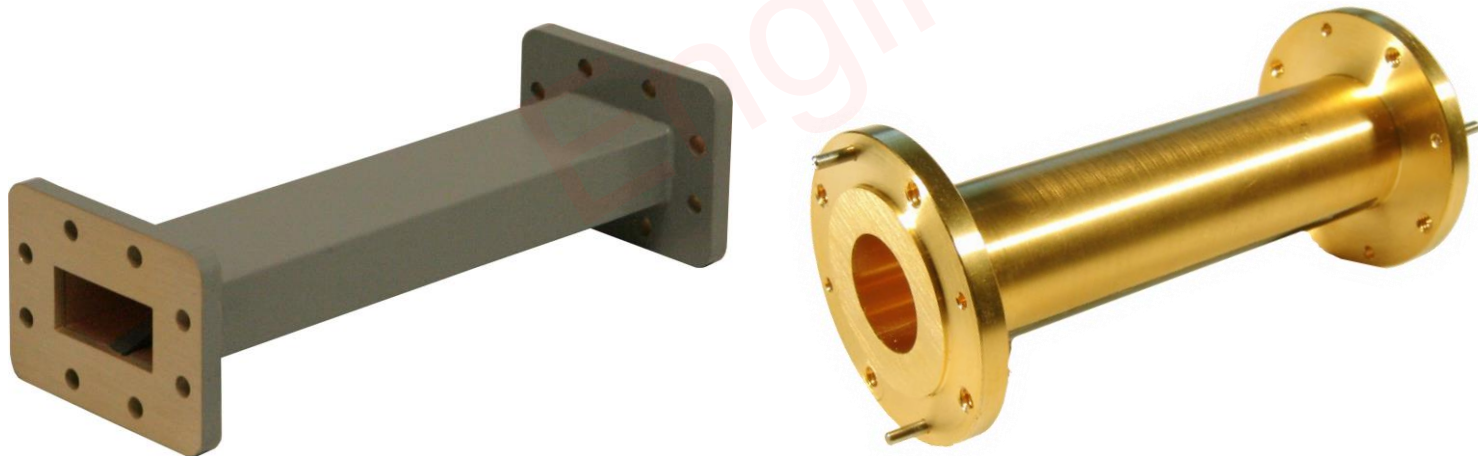


Horn Antenna

Feed
Antenna

Rectangular WG Vs Circular WG

Parameters	Rectangular Waveguide	Circular Waveguide
Filter	❖ It act as High Pass Filter	❖ It act as High Pass Filter
Dominant Mode	❖ TE_{10}	❖ TE_{11}
Mode Analysis	❖ Mode Analysis is Easy.	❖ Mode Analysis is Difficult.
Space	❖ It require less space	❖ It require more space
Fabrication	❖ It isn't easy to manufacture.	❖ It is easy to manufacture.
Waveguide Joining	❖ It is not easy to join two RWGs.	❖ It is easy to join two CWGs.
Tapering Accuracy	❖ Tapering Accuracy is More with RWGs.	❖ Tapering Accuracy is Less with CWGs.
Dispersion	❖ Dispersion is Less with RWGs.	❖ Dispersion is More with CWGs.
Signal Attenuation	❖ Signal Attenuation is Less with RWGs.	❖ Signal Attenuation is More with CWGs.



Examples of Circular Waveguide

Question 1 – The TE_{11} mode propagates through a circular waveguide. The radius of the Guide is 5cm and the guide contains an air dielectric. Determine,

- Cut-off Frequency and Cut-off wavelength
- Phase velocity for an operating frequency of 3GHz
- Guided wavelength
- Wave Impedance

$$\Rightarrow \text{Mode} = TE_{11}$$

$$a = 5\text{cm}$$

$$f = 3\text{GHz} \rightarrow \lambda = \frac{c}{f} = \frac{3 \times 10^8}{3 \times 10^9} = 0.1\text{m}$$

$$\Rightarrow \lambda_c = \frac{2\pi a}{X_{mn}} = \frac{2\pi 5 \times 10^{-2}}{1.841} = 0.1705\text{m}$$

$$\Rightarrow f_c = \frac{c}{\lambda_c} = \frac{3 \times 10^8}{0.1705} = 1.76 \times 10^9 = 1.76\text{GHz}$$

→ $f(3 \text{ GHz}) > f_c(1.76 \text{ GHz})$ [So, 3 GHz signal can propagate through waveguide].

$$\rightarrow v_p = \frac{c}{\sqrt{1 - (f_c/f)^2}} = \frac{3 \times 10^8}{\sqrt{1 - (\frac{1.76}{3})^2}} = \frac{3 \times 10^8}{0.81} = 3.7 \times 10^8 \text{ m/sec}$$

$$\rightarrow \lambda_g = \frac{\lambda}{\sqrt{1 - (f_c/f)^2}} = \frac{0.1}{0.81} = 0.1234 \text{ m}$$

$$\rightarrow (Z_0)_{TE} = \frac{\eta}{\sqrt{1 - (f_c/f)^2}} = \frac{120\pi}{0.81} = 465 \Omega$$

Question 2 – Calculate λ_c , λ_g , f_c , v_p and wave impedance of circular waveguide whose internal diameter is 4.5cm used for a signal at a frequency of 11GHz propagated in TE_{01} mode.

$$\leadsto a = 2.25 \text{ cm}$$

$$f = 11 \text{ GHz} \longrightarrow \lambda = \frac{c}{f} = \frac{3 \times 10^8}{11 \times 10^9} = 0.02727 \text{ m}$$

$$\text{mode} = TE_{01}$$

$$\leadsto \lambda_c = \frac{2\pi a}{X_{m2}} = \frac{2\pi \times 2.25 \times 10^{-2}}{3.83} = 0.0368 \text{ m}$$

$$\leadsto f_c = \frac{v}{\lambda_c} = \frac{3 \times 10^8}{0.0368} = 8.15 \text{ GHz}$$

$$\leadsto f(11 \text{ GHz}) > f_c(8.15 \text{ GHz})$$

[So, 11 GHz signal can propagate through waveguide]

$$\Rightarrow v_p = \frac{c}{\sqrt{1-(\beta/\beta)^2}} = \frac{3 \times 10^8}{\sqrt{1-(\frac{8.15}{11})^2}} = \frac{3 \times 10^8}{0.67} = 4.47 \times 10^8 \text{ m/sec}$$

$$\Rightarrow \lambda_g = \frac{\lambda}{\sqrt{1-(\beta/\beta)^2}} = \frac{0.02727}{0.67} = 0.04 \text{ m}$$

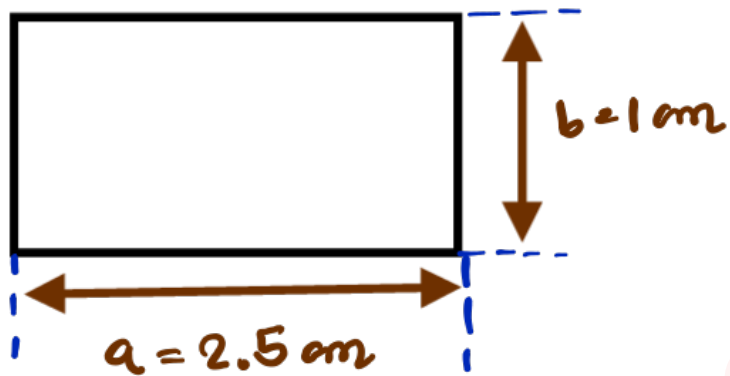
$$\Rightarrow (Z_0)_{TE} = \frac{\eta}{\sqrt{1-(\beta/\beta)^2}} = \frac{120 \pi}{0.67} = 562 \Omega$$

Examples of RWGs

Question 1 – The Dimensions of waveguide is given by $2.5 \times 1 \text{ cm}^2$, The frequency is 8.6GHz.

Find the following:

- Possible Modes
- Cut off Frequency
- Guided Wavelength



$$\rightarrow f_0 = 8.6 \text{ GHz}$$

$$\rightarrow \lambda_0 = \frac{c}{f_0} = \frac{3 \times 10^8}{8.6 \times 10^9}$$
$$= 3.488 \text{ cm}$$

$\rightarrow$ Mode propagation is possible only if

$$\lambda_c > \lambda_0 \quad \text{or} \quad f_c < f_0$$

$$\rightarrow \lambda_c = \frac{2ab}{\sqrt{m^2b^2 + n^2a^2}}$$

$$f_c = \frac{c}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$$

$\rightarrow$ Mode checking Order is

$TE_{10}, TE_{01}, TE_{11},$ TM_{11}	$TE_{21}, TE_{12}, TE_{22}, \dots$ $TM_{21}, TM_{12}, TM_{22}, \dots$
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Compulsory check.

$\Rightarrow \lambda_{cTE_{10}} = 2a = 5 \text{ cm} > \lambda_0$, so TE_{10} mode is possible.

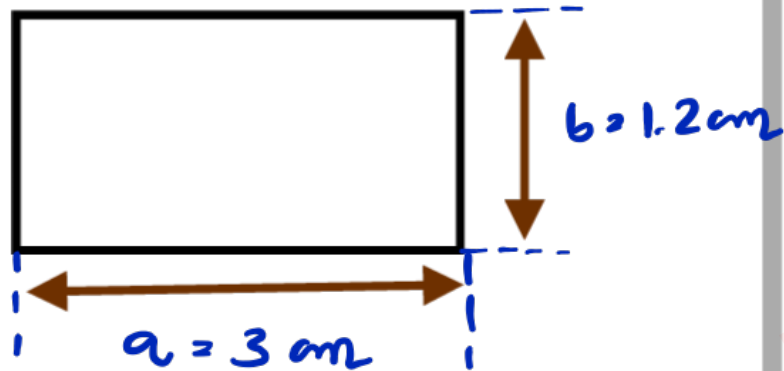
$\Rightarrow \lambda_{cTE_{01}} = 2b = 2 \text{ cm} < \lambda_0$, so TE_{01} mode is not possible.

$\Rightarrow \lambda_{cTE_{11}} = \frac{2ab}{\sqrt{a^2 + b^2}} = \frac{2 \times 2.5 \times 1}{\sqrt{2.5^2 + 1^2}} = 1.85 \text{ cm} < \lambda_0$, so TE_{11} & TM_{11} is not possible.

$$\Rightarrow f_{cTE_{10}} = \frac{v}{\lambda_{cTE_{10}}} = \frac{3 \times 10^8}{5 \times 10^{-2}} = 6 \text{ GHz}$$

$$\Rightarrow \lambda_g = \frac{\lambda_0}{\sqrt{1 - \left(\frac{f_c}{f_0}\right)^2}} = \frac{3.488}{\sqrt{1 - \left(\frac{6}{8.8}\right)^2}} = 4.868 \text{ cm}$$

Question 2 – A Rectangular Metal Waveguide filled with a dielectric material of relative permittivity 4 has the inside dimensions $3 \times 1.2 \text{ cm}^2$. The cut-off frequency for the dominant mode is



→ $\epsilon_r = 4$

→ For dominant mode (TE_{10})

∴ Cut-off frequency.

$$f_c = \frac{c}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$$

$$= \frac{c}{\sqrt{\epsilon_r} \cdot 2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$$

$$= \frac{3 \times 10^8}{2 \times 2} \sqrt{\left(\frac{1}{3 \times 10^{-2}}\right)^2}$$

$$= 2.5 \text{ GHz}$$

Question 3 – The Cut-off frequency of the dominant mode of rectangular waveguide having an aspect ratio of more than 2 is 10GHz. The inner broad wall dimension is given by1.5 cm....

$$\leadsto \frac{a}{b} > 2$$

$$\leadsto f_{cTE_{10}} = 10 \text{ GHz}$$

$$\leadsto a = ?$$

$$\leadsto f_c = \frac{c}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$$

$$\Rightarrow 10^{10} = \frac{3 \times 10^8}{2} \left(\frac{1}{a} \right)$$

$$\Rightarrow a = 1.5 \times 10^{-2} \text{ m}$$

$$\Rightarrow a = 1.5 \text{ cm}$$

Examples of Waveguide

Question 1 – An Air filled Rectangular waveguide has inner dimension of $3 \times 2 \text{ cm}^2$. The wave impedance of TE_{20} mode of propagation in the waveguide at a frequency of 30 GHz is 400Ω [Free Space impedance is given by 377Ω]

$$\Rightarrow a = 3 \text{ cm}$$

$$b = 2 \text{ cm}$$

$$\text{Mode} = \text{TE}_{20}$$

$$f_0 = 30 \text{ GHz}$$

$$\eta = 377 \Omega$$

$$\Rightarrow f_c = \frac{v}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$$

$$= \frac{3 \times 10^8}{2} \left(\frac{2}{3 \times 10^{-2}} \right)$$

$$= 10 \text{ GHz}$$

$$\Rightarrow (Z_w)_{\text{TE}} = \frac{\eta}{\sqrt{1 - (f_c/f_0)^2}}$$

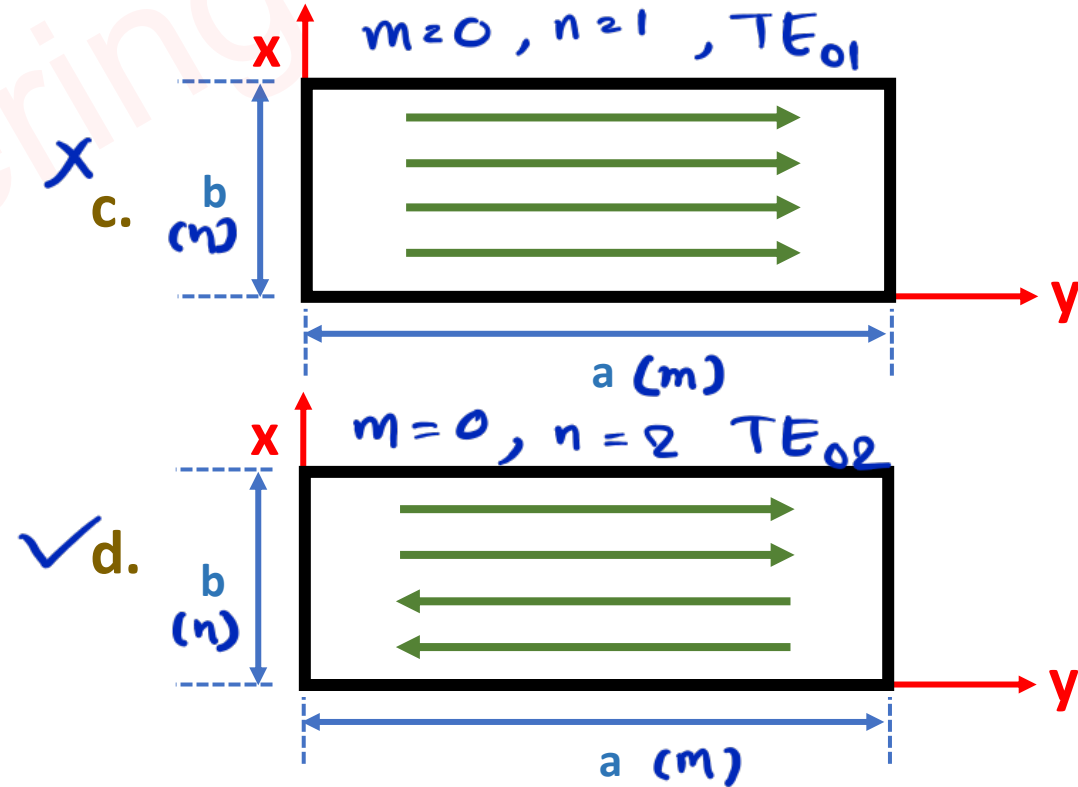
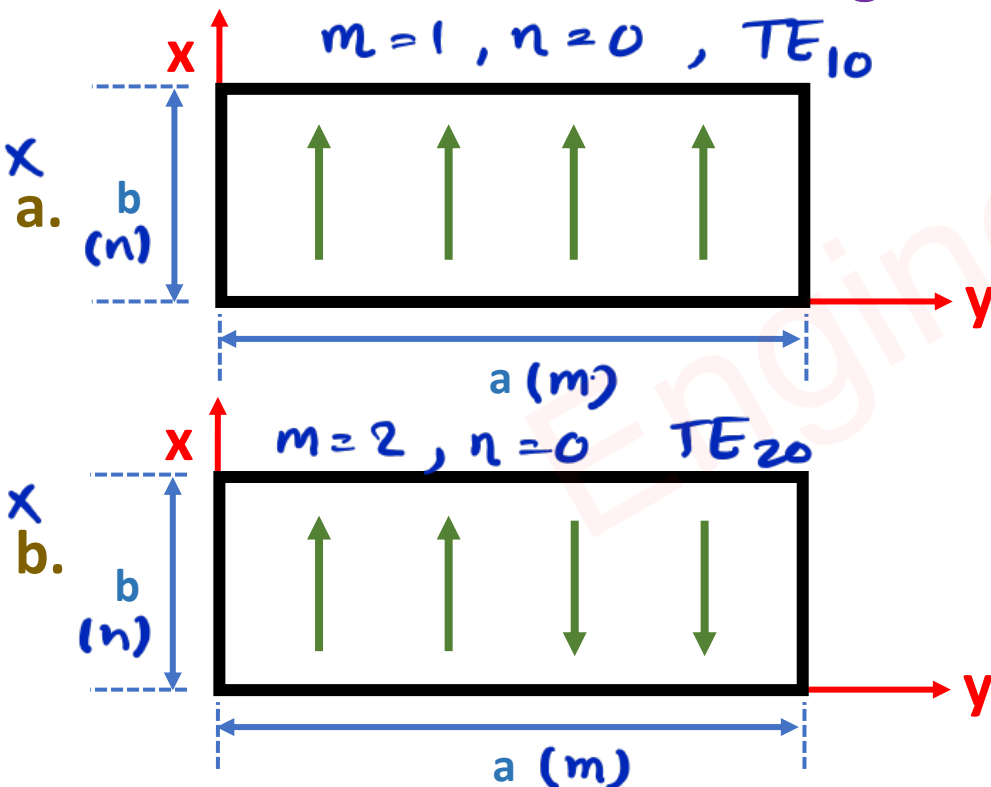
$$= \frac{377}{\sqrt{1 - (10/30)^2}}$$

$$= 400 \Omega$$

Question 2 – Assuming Perfect conductors of transmission line, pure TEM propagation is not possible in

- ☒ a. Co-axial Cable [TEM mode possible]
- ☒ b. Air-Filled Cylindrical Waveguide
- ☒ c. Parallel-twin wireline
- ☒ d. Semi-Infinite Parallel Plate Waveguide

Question 3 – Which of the following represent electric field line for TE_{02} mode in cross-section of hollow metallic waveguide?



Question 4 – Two Rectangular Waveguides, one is air-filled with internal dimensions of $1 \times 2 \text{ cm}^2$ and the other one is dielectric-filled with internal dimension of $1 \times 0.5 \text{ cm}^2$ have the same cut-off frequency for the dominant mode, the relative permittivity of the dielectric material is1.....

(TE₁₀ Mode)

$$\Rightarrow f_c = \frac{c}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2} = \frac{c}{\sqrt{\epsilon_r} 2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$$

$\Rightarrow$ For air-filled waveguide

$$\Rightarrow f_c = \frac{c}{2a} = \frac{c}{2 \times 10^{-2}} \quad \text{--- ①}$$

$\Rightarrow$ For dielectric-filled waveguide

$$\Rightarrow f_c = \frac{c}{\sqrt{\epsilon_r} 2} \times \frac{1}{10^{-2}} \quad \text{--- ②}$$

$\Rightarrow$ Hence, ① = ②

$$\Rightarrow \frac{c}{2 \times 10^{-2}} = \frac{c}{\sqrt{\epsilon_r} \times 2 \times 10^{-2}}$$

$$\Rightarrow \boxed{\epsilon_r = 1}$$

Examples of Waveguide

Question 5 – An Air filled Rectangular waveguide has inner dimension of $a \times b$ cm² ($a > b$). It has cut-off frequency of 6GHz for TE₁₀. For the same waveguide, if the cut-off frequency of TE₁₁ mode is 15GHz, the cut-off frequency for TE₀₁ mode in GHz is

$$\sim f_{c(TE_{10})} = 6 \text{ GHz}$$

$$f_{c(TE_{11})} = 15 \text{ GHz}$$

$$\sim f_c = \frac{c}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$$

$$\sim f_{c(TE_{10})} = \frac{3 \times 10^8}{2a} = 6 \times 10^9$$
$$\Rightarrow a = \frac{1}{40} = 2.5 \times 10^{-2} \text{ m}$$

$$\sim f_{c(TE_{11})} = \frac{3 \times 10^8}{2} \sqrt{\left(\frac{1}{2.5 \times 10^{-2}}\right)^2 + \left(\frac{1}{b}\right)^2}$$

$$\Rightarrow 15 \times 10^9 = 1.5 \times 10^8 \sqrt{1600 + \frac{1}{b^2}}$$

$$\Rightarrow 10000 = 1600 + \frac{1}{b^2}$$

$$\Rightarrow b = 1.09 \times 10^{-2} \text{ m}$$

$$\sim f_{c(TE_{01})} = \frac{3 \times 10^8}{2 \times 1.09 \times 10^{-2}}$$
$$= 13.7 \text{ GHz}$$

Question 6 – The $\vec{E}$ field in a rectangular waveguide of inner dimensions $a \times b$ is given by,

$$\vec{E} = \frac{\omega\mu}{h^2} \left(\frac{\pi}{a}\right) H_0 \sin\left(\frac{2\pi x}{a}\right) \sin(\omega t - \beta z) \quad (E_z = 0)$$

↑ Propagation in z direction

Where, H_0 is a constant, a and b are dimensions along the x-axis and y-axis, respectively. The mode of propagation in the waveguide is

✓ TE_{20} $\leadsto TE_{mn}$ mode

b. TE_{10}

~~a~~ TM_{20}

~~d~~ TM_{11}

$$\leadsto H_z = H_0 \cos\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right)$$

$$\leadsto E_y = \boxed{-\frac{\gamma}{h^2} \frac{\partial E_z}{\partial y}} + j \frac{\omega\mu}{h^2} \frac{\partial H_z}{\partial x}$$

$$= -j \frac{\omega\mu}{h^2} H_0 \underbrace{\cos\left(\frac{n\pi y}{b}\right)}_{\cos 0 = 1} \underbrace{\sin\left(\frac{m\pi x}{a}\right)}_{\boxed{m=2}} \left(\frac{m\pi}{a}\right)$$

$$\boxed{n=0}$$

✓ $\leadsto TE_{20}$

Question 7 – For a normal mode EM wave Propagation in a hollow rectangular waveguide:

- a✓ The Phase velocity is greater than the group velocity
- b✓ The phase velocity is greater than velocity of light in free space
- ~~c~~ The phase velocity is less than the velocity of light in free space
- ~~d~~ The phase velocity may be either greater than or less than the group velocity

$$\Rightarrow v_p = \frac{c}{\sqrt{1 - (f_c/f_0)^2}} \quad \sim \quad v_g = c \sqrt{1 - (f_c/f_0)^2}$$

⇒ For Normal mode EM wave cond.ⁿ is

$$f_c < f_0$$

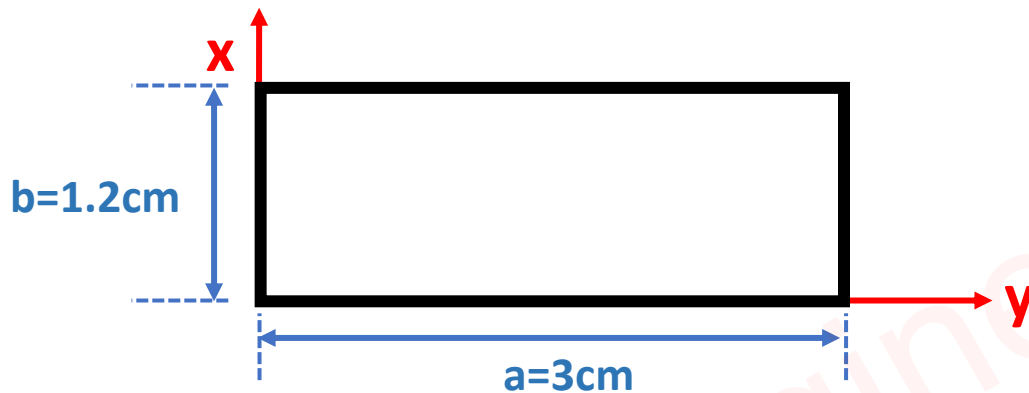
$$\Rightarrow v_p > c > v_g$$

Examples of Waveguide

Question 8 – The magnetic field along the propagation direction inside a rectangular waveguide with the cross-section shown in the figure is

$$H_z = 3 \cos(2.094 \times 10^2 x) \cos(2.618 \times 10^2 y) \cos(6.28 \times 10^{10} t - \beta z)$$

$$H_z = H_0 \cos\left(\frac{m\pi}{a} x\right) \cos\left(\frac{n\pi}{b} y\right) \cos(\omega t - \beta z)$$



$$\sim \omega = 6.28 \times 10^{10} \text{ Rad/sec} \Rightarrow f_0 = \frac{\omega}{2\pi} = 10 \text{ GHz}$$

$$\sim \frac{m\pi}{a} = 2.094 \times 10^2 \Rightarrow \frac{m}{a} = 66.5$$

$$\sim \frac{n\pi}{b} = 2.618 \times 10^2 \Rightarrow \frac{n}{b} = 83.3$$

The phase velocity v_p of the wave inside the waveguide satisfies

a. $v_p > c$

b. $v_p = c$

☒ c. $v_p = 0$

d. $0 < v_p < c$

$$\begin{aligned} \sim f_c &= \frac{c}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2} \\ &= \frac{3 \times 10^8}{2} \sqrt{66.5^2 + 83.3^2} \\ &= 16 \text{ GHz} \end{aligned}$$

$\sim f_c > f_0$, means EM waves can not propagate through RWG.

$$\boxed{v_p = 0}$$

Question 9 – For a rectangular waveguide of internal dimensions $a \times b$ ($a > b$), the cut-off frequency for the TE_{11} mode is the arithmetic mean of the cut-off frequencies for TE_{10} mode and TE_{20} mode. If $a = \sqrt{5}$ cm, the value of b (in cm) is2 cm.....

$$\Rightarrow f_c = \frac{c}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$$

$$\Rightarrow f_{c(TE_{11})} = \frac{f_{c(TE_{10})} + f_{c(TE_{20})}}{2}$$

$$\Rightarrow \sqrt{\frac{1}{a^2} + \frac{1}{b^2}} = \frac{\frac{1}{a} + \frac{2}{a}}{2}$$

$$\Rightarrow \sqrt{\frac{1}{a^2} + \frac{1}{b^2}} = \frac{3}{2a}$$

$$\Rightarrow \frac{1}{a^2} + \frac{1}{b^2} = \frac{9}{4a^2}$$

$$\Rightarrow \frac{1}{5} + \frac{1}{b^2} = \frac{9}{20}$$

$$\Rightarrow \frac{1}{b^2} = \frac{9-4}{20}$$

$$\Rightarrow \frac{1}{b^2} = \frac{1}{4}$$

$$\Rightarrow \boxed{b = 2 \text{ cm}}$$

Question 10 – For a Rectangular Waveguide with dimension $a = 2.29\text{cm}$ and $b = 1.02\text{cm}$. Mode of operation is TE_{10} . The range of operating frequency is in such a way, that the frequency must be 25% higher than the cut-off frequency and not greater than 95% of the next higher cut-off frequency.

a. $6.55\text{GHz} < f < 13.1\text{GHz}$

✓ b. $8.19\text{GHz} < f < 12.45\text{GHz}$

c. $1.64\text{GHz} < f < 10.24\text{GHz}$

d. $8.19\text{GHz} < f < 13.1\text{GHz}$

$\rightarrow a = 2.29\text{ cm}$
 $b = 1.02\text{ cm}$ } Here a is more than double of b .

$f_0 > 25\% \text{ Higher } (\text{TE}_{10} \text{ mode})$

$f_0 < 95\% \text{ lower } (\text{TE}_{20} \text{ mode})$

$\rightarrow f_c = \frac{c}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$

$\rightarrow f_{c(\text{TE}_{10})} = \frac{c}{2a} = 6.55\text{ GHz}$

$f_{c(\text{TE}_{20})} = \frac{c}{2} \times \frac{2}{a} = 13.1\text{ GHz}$

$\Rightarrow 1.25 f_{c(\text{TE}_{10})} > f_0 > 0.95 f_{c(\text{TE}_{20})}$

$\Rightarrow 1.25 \times 6.55 > f_0 > 0.95 \times 13.1$

$\Rightarrow 8.19\text{ GHz} > f_0 > 12.45\text{ GHz}$